

Code-Layout, Readability and Reusability

Date	2 July 2026
Team ID	XXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S. No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent indentation maintained
2	Proper File Structure	Files and folders are logically organized	Yes	Proper folder structure followed
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Meaningful variable names used
4	Function / Method Names	Functions are descriptively named	Yes	Descriptive function names used
5	Code Comments	Inline and block comments explain log	Partial	Limited code comments added
6	Modular Design	Code is split into reusable functions/m	Yes	Modular design implemented
7	No Redundant Code	Duplicate or unused code is removed	Yes	Unused code removed
8	Error Handling	Exceptions and errors are handled gradually	Yes	Error handling implemented

Reusable Components / Modules:

S. No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Crop Prediction Module	Python, scikit-learn	Crop Recommendation	High

2	Input Validation Module	Python, JavaScript	Form Validation	High
3	Flask Backend Module	Python, Flask	Web Application Services	High
4	Model Loading Module	Python, Pickle	Prediction Engine	High
5	UI Components	HTML, CSS, JavaScript	Frontend Pages	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure
Readability	4	Easy to understand and maintain
Reusability	5	Reusable modules implemented
Documentation / Comments	4	Basic documentation provided
Overall Score	5	Good quality and maintainable code